

**Mahler's 10th symphony and shape of dissonance:**

**An exploration in acoustics and dissonance**

Veronica Naselli

Albert Clement

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## Introduction

Although for most people the ability to distinguish a dissonant sound is innate, perhaps rarely do they ask themselves the question of why certain sounds are perceived as dissonant by the human cognition. Indeed, the ability to perceive different sounds underlies the existence of music: it is something so essential that it is accepted with no second thought. At times, however, we encounter something striking enough that we start to question why it works, and such is the case with Gustav Mahler's 10th symphony. The symphony is a work in dissonance and its effects on the human perception: Mahler utilizes sound to express his despair and devastation to such proficiency, that the performers of this symphony seem to produce human screams with their instruments. Although much could be said about this composition, thick with meaning and unfinished history, I will focus on one exemplary moment, the 9-notes, altered dominant chord in F# major at the climax of the first movement. I am going to investigate how the chord fits within the piece's narrative and Mahler's own life, and furthermore why it sounds remarkably dissonant by exploring music theory and current acoustics theories.

### Part 1: The narrative as reflection of his life

To investigate the impact that such a chord has on the listener, it is best to contextualize the symphony within Mahler's life and therefore explain the stylistic and acoustic choices made by the composer. The composition's storyline delineates a tale of heartbreak and desperation, following his beloved wife Alma's affair with the young architect Walter

Gropius and the diagnosis of a heart condition in 1907.<sup>1</sup> This is suggested by the annotations in the original manuscript itself, in which Mahler writes: ‘The Devil dances it with me / Madness take me, cursed one! / Destroy me, that I forget that I am! / that I cease to be, that I de... [das ich ver... ]’, and ”You alone know what it means! ... Ah! Farewell my lyre!”. Further on, he calls out to her by her affectionate nickname, Almschi, preceded by a moving declaration of his devotion: “To live for you! To die for you!”.<sup>2</sup> Among his many achievements, his marriage to Alma was arguably the most significant in influencing the composition of this piece. As we can gather from the clues given to us in the manuscript and from the context given by other events in his life, his feelings for her ran deep, although reportedly not smoothly. The turbulence in his family life, caused by the death of his younger daughter Maria in 1907,<sup>3</sup> his failing heart, and Alma’s affair, caused a great upheaval in his life. Although aware of his fragile health, he insisted on carrying out his previous commitments and up until his death in 1910, he worked on composition and lead conducting tours across Europe and the US.<sup>4</sup> As it happens, he was working on the 10th symphony when he died in 1911 in Vienna, after contracting bacterial endocarditis.<sup>5</sup>

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1. Peter Franklin, “Mahler, Gustav” [in en], in *Oxford Music Online* (Oxford University Press, 2001), accessed November 29, 2023, <https://doi.org/10.1093/gmo/9781561592630.article.40696>, <http://www.oxfordmusiconline.com/grovemusic/documentID/omo-9781561592630-e-0000040696>.

2. Franklin.

3. Franklin.

4. Franklin.

5. Franklin.

The manuscript was left unfinished in 1910, which prompted later composers to take up the mantle of arranging it for orchestra performance. The work was fully sketched out in short score but was only orchestrated in the first movement and the first measures of the third,<sup>6</sup> and the short score presented several inconsistencies and imperfections which proved very challenging for posthumous interpreters.<sup>7</sup> The history of the adaptation of the symphony is quite intricate, as multiple people worked on it at different times, and Alma Mahler herself guarded the manuscript for twelve years, before allowing a facsimile copy to be made in 1924.<sup>8</sup> The ethics of completing Mahler's work after his express wish against it before his death raised some eyebrows, especially in the case of larger additions, such as Deryck Cooke's rendition. Some of the people who attempted to complete the extant material are, in order, Ernst Krenek, Franz Schalk, Willem Mengelberg, and Alexander Zemlinsky.<sup>9</sup> Krenek was the first who put the first two movements into performing arrangement, since they were left mostly finished, and each following attempt contributed a small part to the symphony in order for it to be performed.<sup>10</sup> None of them had yet attempted to finish the last movements, which were left with gaps and largely in sketch form. In the case of Krenek's version, he has denied responsibility and attributed the addition to Franz Schalk, under which the Adagio

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6. Michael Steinberg, "Symphony No. 10 by Gustav Mahler" [in en].

7. Deryck Cooke, "Mahler's Tenth Symphony: Artistic Morality and Musical Reality," *The Musical Times* 102, no. 1420 (June 1961): 351, accessed November 29, 2023, <https://doi.org/10.2307/950812>, <https://www.jstor.org/stable/950812?origin=crossref>.

8. Frans Bouwman, "Editing Mahler 10: Unfinished Business," Publisher: Musical Times Publications Ltd. *The Musical Times* 142, no. 1877 (2001): 43–51, accessed November 29, 2023, <https://doi.org/10.2307/1004576>, <https://www.jstor.org/stable/1004576>.

9. Bouwman.

10. Bouwman.

and *Purgatorio* were performed in 1924, in Vienna.<sup>11</sup> The responsibility to finish another composer's work felt too heavy for most composers of the time, and several conductors such as Leonard Bernstein, Bruno Walter, and Pierre Boulez refused to perform these earlier attempts, possibly due to the patchiness of the finished product.<sup>12</sup> Deryck Cooke has made a celebrated attempt, which is often performed by those who wish to bring the composition to the public. He is not the only one who attempted it, but he has created a widely accepted reconstruction, which is no small feat, by completing all movements on the basis of the facsimile of the manuscript published in 1924.<sup>13</sup> Cooke has defended his position in the 1961 article "Mahler's Tenth Symphony: Artistic Morality and Musical Reality", published in *The Musical Times*, where he asserts that he respects Mrs. Mahler's decision of not allowing any more performances, and that although the Tenth Symphony was lacking significant parts and was not viable to be performed as left in the manuscript, it was fully sketched out, and therefore he, along with anybody else who might try to fill in the gaps, was working off a completed work, at least in its conceptualization.<sup>14</sup> His work, due to its quality and the integrity, became the standard for recordings and performances. His intention, as he himself stated, was not to complete Mahler's work as he would have done it, as that would have been an impossible endeavour, but rather to offer a version of it which could be performed to the

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11. Cooke, "Mahler's Tenth Symphony."

12. Steinberg, "Symphony No. 10 by Gustav Mahler."

13. Cooke, "Mahler's Tenth Symphony."

14. Cooke.

public.<sup>15</sup> Whichever version one might choose to regard as the closest to Mahler's original intention, we are lucky to have it be passed on to us at all, let alone for it to be performed in any way close to complete. Other attempts were made before him and after him, but for the purpose of this paper I will choose to follow later renditions as little as possible, and if needed, I will refer to Cooke's rendition, in light of its popularity and completeness.

In the interest of my exploration, I will choose to focus on the first movement of the symphony, the Adagio. One chord at the climax of the movement expresses the height of the pain felt by Mahler in the most indicative way. Towards the end of the Adagio, trumpet seems to scream in human pain, and the orchestra wants to drown it through a series of dissonant chords,<sup>16</sup> among which there is the altered F# dominant chord I will take into consideration. The chord in question contains the following pitches: C#2 - G#2 - B2 - D3 - F3 - A5 - C6 - Eb6 - G6 .

From the beginning of the Adagio, the violas begin with a tentative, wandering tune, bringing the audience into a state of uneasy and somewhat muted melancholy. As the Adagio proper then begins, the strings and the woodwinds paint a picture of building tragedy, with a low, building bass, and rising shrill screams from the violins and the wind instruments. The violas carry the undercurrent of doubt, and the violins along with the winds build the wave of feeling that Mahler was feeling. There is no doubt the narrative of the piece now is great, unequivocal heartbreak and tragedy, of a love betrayed. The high-pitched violin and

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15. Theodore Bloomfield, "In Search of Mahler's Tenth: The Four Performing Versions as Seen by a Conductor," Publisher: Oxford University Press, *The Musical Quarterly* 74, no. 2 (1990): 175–196, accessed November 29, 2023, <https://www.jstor.org/stable/742188>.

16. Steinberg, "Symphony No. 10 by Gustav Mahler."

winds give the impression of a human scream, as they take charge in a wave of dissonant voices. As such, the altered chord this paper is going to examine expresses the peak of his suffering through extreme dissonance, and therefore it lands itself well to the purpose of this paper.

## Part 2: The shape of sound

To investigate the reason why this chord sounds dissonant, we need to look into the field of acoustics. Acoustics studies the phenomenon of sound and all its properties, among which there is frequency. The tools from physics theory can help us visualize dissonance as concrete numbers. The first step will be to compare the ratios between the frequencies of the pitches, and the second step will be to explore their dissonance from both a musical and a physical standpoint.

To establish the correct frequencies for each pitch, we need to establish the correct framework. Mahler probably worked within an equal temperament framework, and this assumption can be supported if one thinks of when the widespread use of the equal temperament system began, which dates to the 19th century.<sup>17</sup> Pitch is relative and the concept has undergone variation across the centuries, so it is important to establish the correct temperament because it informs us on the ratio of frequencies between pitches. The equal temperament system was originally developed as a compromise tuning scheme, so that each step in the chromatic scale would be, as the name suggests, equal. The system works simply, as the

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17. Burkholder Peter J., Grout Jay Donald, and Palisca V Claude, *A History of Western Music: Tenth International Student Edition* [in en], Google-Books-ID: 15ibDwAAQBAJ (W.W. Norton & Company, February 2019).

octave is divided in twelve parts, so that the ratios for each interval contained in it approach 2:1. To achieve such a result, the twelfth root of 2 (1.0594630944) is multiplied by itself 12 times, which approximates 2 very closely. The only two intervals which are perfectly tuned within the equal temperament are the unison (1:1) and the octave (2:1), while all other intervals are results of compromise, as they need to be minutely adjusted in order for the negligible difference in beats to be unnoticeable and for each step in the chromatic scale to appear identical.<sup>18</sup> The cents system was introduced in 1884 by J. Ellis, who developed a system by which a factor of  $\sqrt[1200]{2}$  was used to produce 1200 intervals within an octave. A value  $c$  was then established, such that there could be an approximation of the interval between two frequencies, and a value  $n$ , corresponding to the frequency ratio of the interval.  $c$  was obtained through the following formula:

$$c = 1200 \times \frac{\log n}{\log 2} \quad (1)$$

Such an approximation allows for much easier calculations and simpler numbers denoting the ratios. The corresponding cents approximations will also be included in Table 1. Each sound has a frequency, which is measured with the unit Hertz, defined as follows:

$$\text{Hz} = \frac{1}{s} \quad (2)$$

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18. John W. Link, "Understanding the Two Great Temperaments: Equal and Meantone," Publisher: [MENC: The National Association for Music Education, Sage Publications, Inc.] *Journal of Research in Music Education* 13, no. 3 (1965): 136–146, accessed December 30, 2023, <https://doi.org/10.2307/3343668>, <https://www.jstor.org/stable/3343668>.

.<sup>19</sup> The Hertz unit of measurement is used to measure any phenomenon with regular periodic variations, most commonly for sound, electromagnetic waves, and alternating electric currents. In the interest of this investigation, we will assume all sound taken into consideration is a sinusoidal phenomenon.<sup>20</sup> Any system of temperament and tuning works off of a base note reference upon which all relative intervals are built. The frequencies of pitches tuned to equal temperament commonly have the reference note A, measuring 440 Hz.<sup>21</sup> Therefore, for the notes C# - G# - B - D - F - A - C - Eb - G, we can deduce their frequencies by first identifying the intervals between the reference A and each pitch, then applying the corresponding chromatic distance to the Hz value, thus obtaining the Hz value for each frequency. All intervals and corresponding semitone distances are included in Table 1 (1).

By applying the interval ratios in equal temperament with a reference of A above middle C as 440 Hz, the frequency of C4 is calculated to be about 277.183 Hz. Equation (3) is therefore the formula used to obtain frequency  $f_n$ , for fundamental frequency  $f_0$ , twelfth-root of 2  $r$ , and  $n$  of semitones between the notes:

$$f_n = f_0 \times r^n \quad (3)$$

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19. *Hertz — Definition & Facts — Britannica* [in en], accessed December 26, 2023, <https://www.britannica.com/science/hertz>.

20. Thomas Rossing, *Springer Handbook of Acoustics* [in en], Google-Books-ID: oWExBgAAQBAJ (Springer, January 2015).

21. Rossing.

| Equal temperament |            |       |                                  |
|-------------------|------------|-------|----------------------------------|
| Interval          | Ratio      | Cents | Semitone<br>for each<br>interval |
| Unison            | 1:1        | 0     | 0                                |
| Minor Second      | 1.059463:1 | 100   | 1                                |
| Major Second      | 1.122462:1 | 200   | 2                                |
| Minor Third       | 1.189207:1 | 300   | 3                                |
| Major third       | 1.259921:1 | 400   | 4                                |
| Perfect Fourth    | 1.334840:1 | 500   | 5                                |
| Tritone           | 1.414214:1 | 600   | 6                                |
| Perfect Fifth     | 1.498307:1 | 700   | 7                                |
| Minor Sixth       | 1.587401:1 | 800   | 8                                |
| Major Sixth       | 1.681793:1 | 900   | 9                                |
| Minor Seventh     | 1.781797:1 | 1000  | 10                               |
| Major Seventh     | 1.887749:1 | 1100  | 11                               |
| Octave            | 2:1        | 1200  | 12                               |

Table 1. Table with main intervals and corresponding ratios and cents in the equal temperament system.

As an example, we will evaluate A5's frequency, present in the chord, step by step:

First, let us establish that A4's fundamental frequency is set at  $440Hz$  and that A4 and A5 are one octave apart. Then, let us remember that the semitones in an octave are 12. We can thus plug in the values in the formula:

$$f = 440Hz \times 1.0594630944^{12} \quad (4)$$

This equation will result in:

$$f = 880Hz \quad (5)$$

Table 2 will show all the results of the above calculations, and each pitch will be reported with letter name and number. In the case of compound intervals, the representation will

be constituted by a two-part definition of octave(s) and remaining interval. The chromatic distance that will be associated will be the one corresponding to the remaining interval, given that each compound interval possesses the same qualities as its simple counterpart.<sup>22</sup> A negative sign indicates that the pitch is below A4, while a positive sign indicates it is above A4.

Music theory teaches us that certain intervals are consonant and others are dissonant, based

| Pitch and Relative Frequency |                                   |           |            |
|------------------------------|-----------------------------------|-----------|------------|
| Pitch                        | Interval from A4                  | Semitones | Frequency  |
| C#2                          | Two octaves and a minor sixth     | -8        | 69.30 Hz   |
| G#2                          | Two octaves and a minor second    | -1        | 103.83 Hz  |
| B2                           | One octave and a minor seventh    | -10       | 123.47 Hz  |
| D3                           | One octave and a perfect fifth    | -7        | 146.83 Hz  |
| F3                           | One octave and a major third      | -4        | 174.61 Hz  |
| A5                           | One octave                        | +12       | 880 Hz     |
| C6                           | One octave and a minor third      | +3        | 1046.50 Hz |
| Eb6                          | One octave and a diminished fifth | +6        | 1396.91 Hz |
| G6                           | One octave and a minor seventh    | +10       | 1567.98 Hz |

Table 2. Table with pitches and corresponding ratios and frequencies in the equal temperament system.

on conventional agreements on the physical perception of such sounds, explained by many theories across the history of Western music. The theory offers an interpretation for the

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22. Peter, Jay, and V, *A History of Western Music*.

discordancy of the chord, which will serve as a base for our physical proof. In Table 3, the adjacent intervals within the chord are shown, while in Table 4 all intervals from the root are shown. Note that in the case of compound intervals, the corresponding simple intervals are shown.

| Adjacent intervals within altered F# V chord |                 |
|----------------------------------------------|-----------------|
| Pitches                                      | Intervals       |
| C#2 - G#2                                    | Perfect fifth   |
| G#2 - B2                                     | Minor third     |
| B2 - D3                                      | Minor Third     |
| D3 - F3                                      | Minor third     |
| F3 - A5                                      | Augmented sixth |
| A5 - C6                                      | Minor Third     |
| C6 - Eb6                                     | Minor Third     |
| Eb6 - G6                                     | Major Third     |

Table 3. Adjacent intervals in altered F# dominant chord.

| Intervals from root |               |
|---------------------|---------------|
| Pitches             | Intervals     |
| C#2 - G#2           | Perfect Fifth |
| C#2 - B2            | Minor Sixth   |
| C#2 - D3            | Major Seventh |
| C#2 - F3            | Major Third   |
| C#2 - A5            | Minor Sixth   |
| C#2 - C6            | Major Seventh |
| C#2 - Eb6           | Minor Third   |
| C#2 - G6            | Tritone       |

Table 4. Intervals from root to each pitch in the altered F# dominant chord.

It is clear from 3 that most of the adjacent intervals within the chord are minor thirds, which would usually be considered imperfect consonances, however the presence of so many of them, with the destabilizing influence of the augmented sixth interval, determines a sense of tension and discordance. Table 4 4 also highlights the presence of a major seventh

and a tritone in the intervals from the root C#2, with otherwise imperfect consonances, therefore lacking a balancing perfect consonance. Overall, music theory tells us that this is without a doubt a dissonant chord, which lacks balance and structure. Many theories have been put forward in order to understand the perception of sonance by the human cognition, however the many factors which influence dissonance perception, such as cultural and social norms, and the multitude of neural processes needed to process a sound, cause this complex issue to still be a matter of debate.<sup>23</sup> Helmholtz's theory proposes that sonance is dependent on the difference between the beatings of the partials of each note: dissonance would then depend on both the difference between the fundamental tones and the difference between the overtones of sounds.<sup>24</sup> It is important to note that although we might be able to calculate a numerical indication of sonance, cultural and social teachings influence largely the subjective perception of dissonance and therefore a dual interpretation including subjective aspects must be factored in. In short, Helmholtz argues that when two tones occur in the same acoustical space, and they differ slightly in frequency (pitch), then they will create beats when the two fluctuating amplitudes overlap.<sup>25</sup> To obtain the frequency of the beats, one must subtract the frequencies of the tones. Such a frequency, were it to fall

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23. Phil N. Johnson-Laird, Olivia E. Kang, and Yuan Chang Leong, "On Musical Dissonance," *Music Perception* 30, no. 1 (September 2012): 19–35, accessed January 2, 2024, <https://doi.org/10.1525/mp.2012.30.1.19>, <https://doi.org/10.1525/mp.2012.30.1.19>.

24. Hermann Ludwig Ferdinand von Helmholtz, 1821-1894. Helmholtz, Alexander J. Ellis, and Henry 1901-Margenau, *On the sensations of tone as a physiological basis for the theory of music* [in Inglese], The 2nd English ed., transl., thoroughly rev. and corr. rendered conformal to the 4th (and last) German ed. of 1877, with numerous additional notes and a new additional appendix bringing down information to 1855, and especially adapted to the use of music students by Alexander J. Ellis. With a new introd. (1954) by Henry Margenau. [Repr.], Dover histories, biographies and classics of mathematics and the physical sciences, Section: XXX, 576 p. : ill. ; 24 cm (New York: Dover, 1970).

25. Helmholtz, Ellis, and Margenau.

within a certain range, may give rise to a "jarring" effect, or a dissonance.<sup>26</sup> Therefore, it is possible to determine whether certain frequencies will give cause for dissonance. Another factor that weighs on the determination of sonance are partials: if upper partials create a distance of at least a semitone, a disturbance will be heard, however it is cancelled out if the tones possess the same lower partials.<sup>27</sup> Helmholtz found that tritones are majorly affected by such disturbances, while the minor sixth and major third are only moderately affected, allowing them to be accepted as imperfect consonances.<sup>28</sup> This considerations however must be taken into context, as within a chord, interactions between tones are much more complex than within a dyad, and therefore much more pronounced in the case of jarringly different overtones, such as this case.<sup>29</sup> Due to the considerations above within musical theory and based on the analysis made on the intervals within the chord, we can conclude that this chord is dissonant, both from a musical and physical standpoint.

## Conclusions

Dissonance is a fascinating phenomenon that has been used as a tool to convey a great deal of messages, and by Mahler in particular to articulate desperation and suffering. The aim of this paper was to offer an exploration into a musical feature often given for granted, and to investigate the latest theories on the topic. The paper gave a brief overview of the history behind the composition of the 10th Symphony in order to give context for

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26. Helmholtz, Ellis, and Margenau, *On the sensations of tone as a physiological basis for the theory of music*.

27. Helmholtz, Ellis, and Margenau.

28. Helmholtz, Ellis, and Margenau.

29. Helmholtz, Ellis, and Margenau.

such a message to be conveyed, then went on to focus on one specific chord in the Adagio, to exemplify this phenomenon. Music theory was used to explain the extreme dissonance of the chord taken into consideration, and determined that most of the intervals within the chord are traditionally classified as dissonant. Having proven that, an overview on acoustics theory is brought forward, highlighting the Helmholtz framework as a widely accepted explanation of the perception of dissonance. Although there is not yet a definitive answer as to why dissonance is perceived as such, future researchers who would like to attempt this might have more luck in illuminating such a complex issue in order to further musicologists' understanding of musical phenomena from a physical perspective.

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